

# Bayesian Age-Period-Cohort Modeling

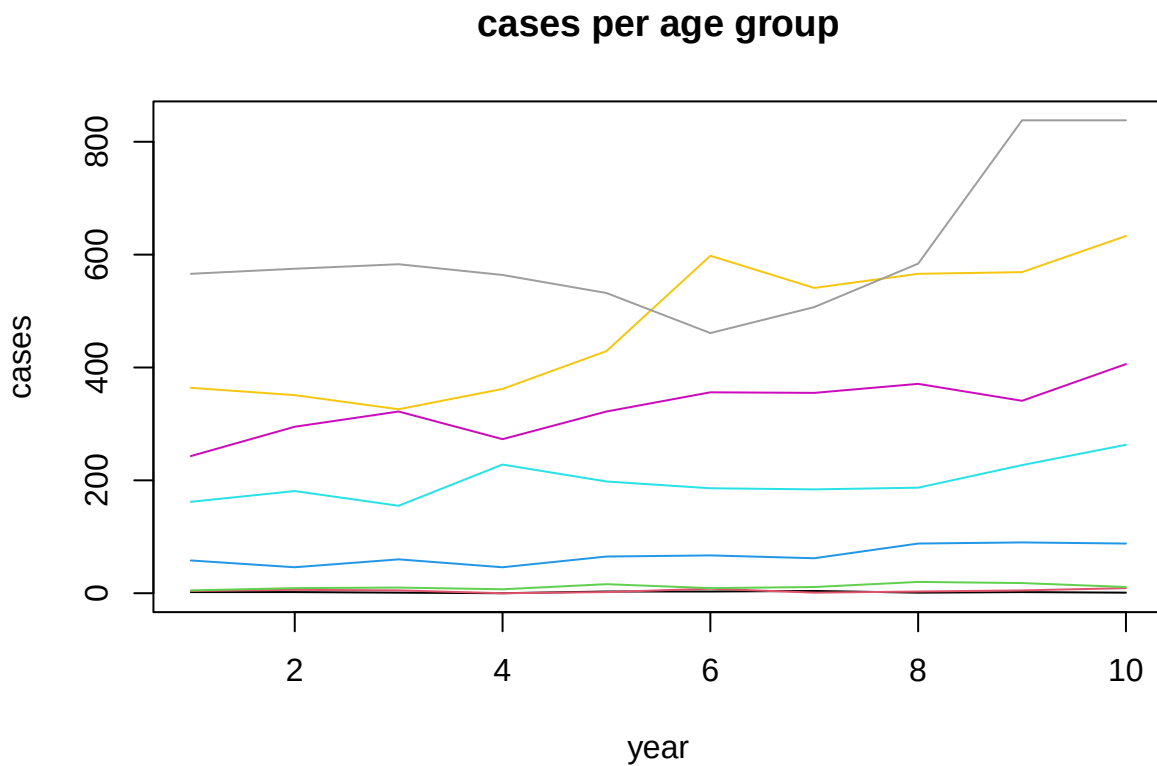
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## Data example

BAMP includes a data example.

```
data(apc)
plot(cases[,1],type="l",ylim=range(cases), ylab="cases", xlab="year", main="cases per age group")
for (i in 2:8)lines(cases[,i], col=i)
```



## APC model with random walk first order prior

```
model11 <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
  periods_per_agegroup = 5)
```

bamp() automatically performs a check for MCMC convergence using Gelman and Rubin's convergence diagnostic. We can manually check the convergence again:

```
checkConvergence(model1)
```

```
## [1] TRUE
```

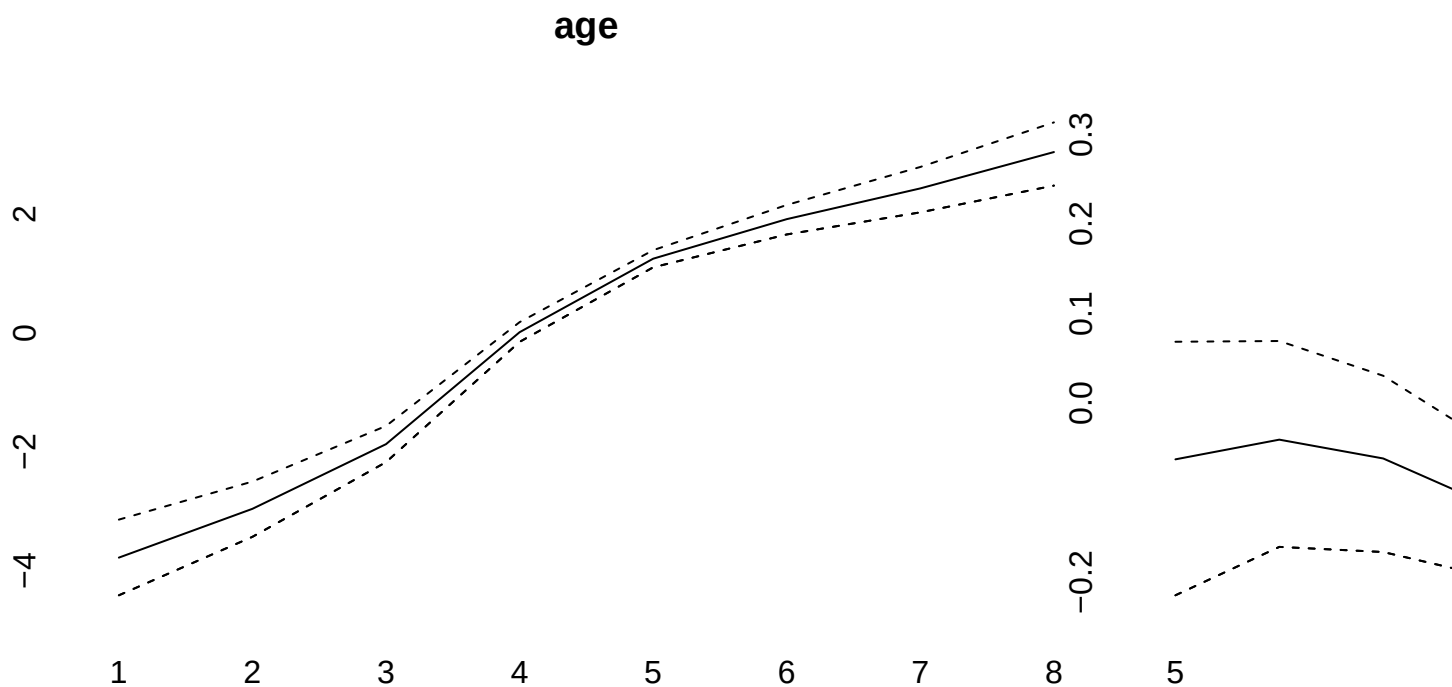
Now we have a look at the model results. This includes estimates of smoothing parameters and deviance and DIC:

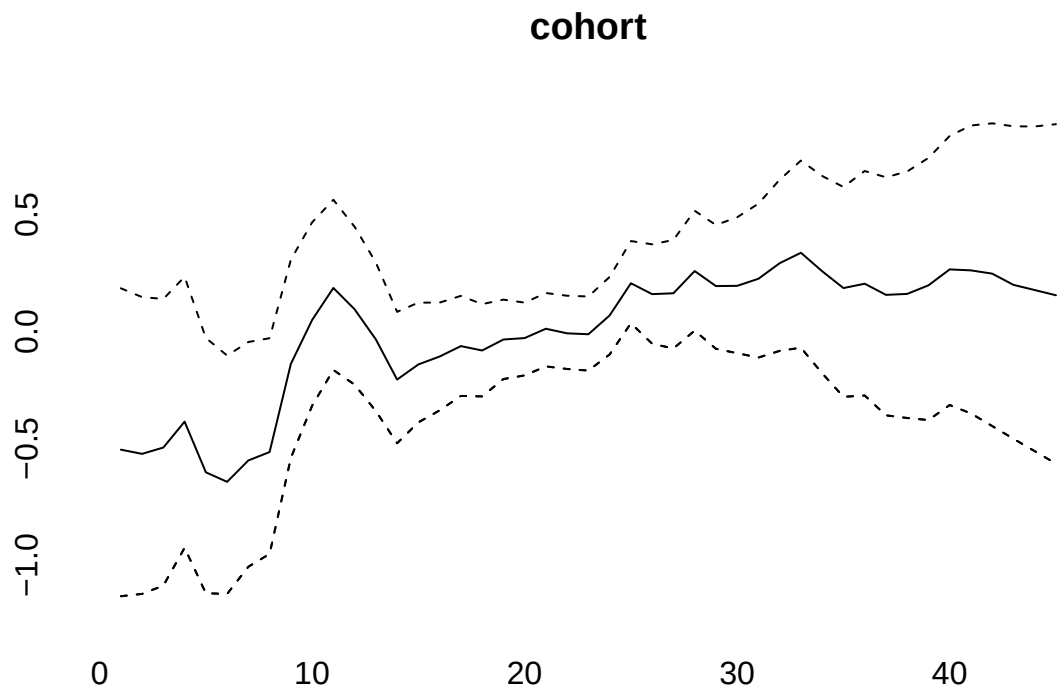
```
print(model1)
```

```
##
## Model:
## age (rw1) - period (rw1) - cohort (rw1) model
## Deviance:      231.42
## pD:            37.10
## DIC:           268.52
##
##
## Hyper parameters:           5%           50%           95%
## age                        0.358        0.925        1.922
## period                     66.574       203.634      612.494
## cohort                     34.331        59.311       96.838
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

We can plot the main APC effects using point-wise quantiles:

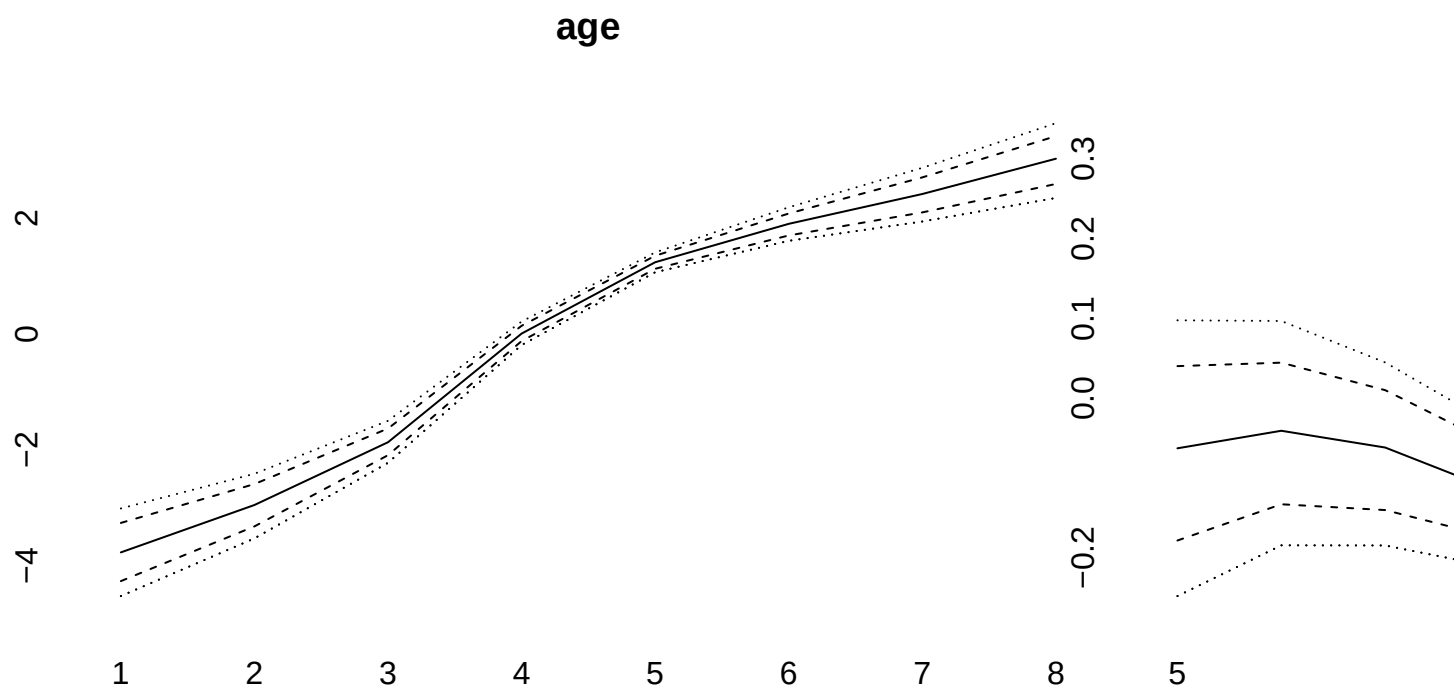
```
plot(model1)
```

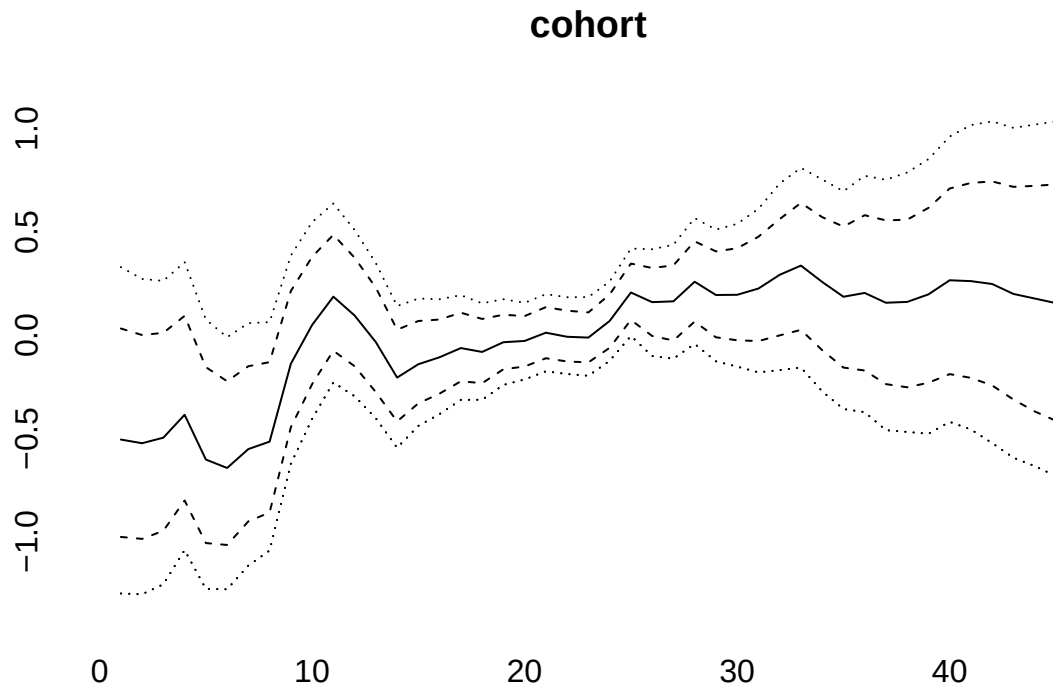




More quantiles are possible:

```
plot(model1, quantiles = c(0.025,0.1,0.5,0.9,0.975))
```





### APC model with random walk second order prior

```
model2 <- bamp(cases, population, age="rw2", period="rw2", cohort="rw2",
  periods_per_agegroup = 5,
  mcmc.options=list("number_of_iterations"=200000, "burn_in"=100000, "step"=50, "tuning"=50000),
  hyperpar=list("age"=c(1,.5), "period"=c(1,0.05), "cohort"=c(1,0.05)))
```

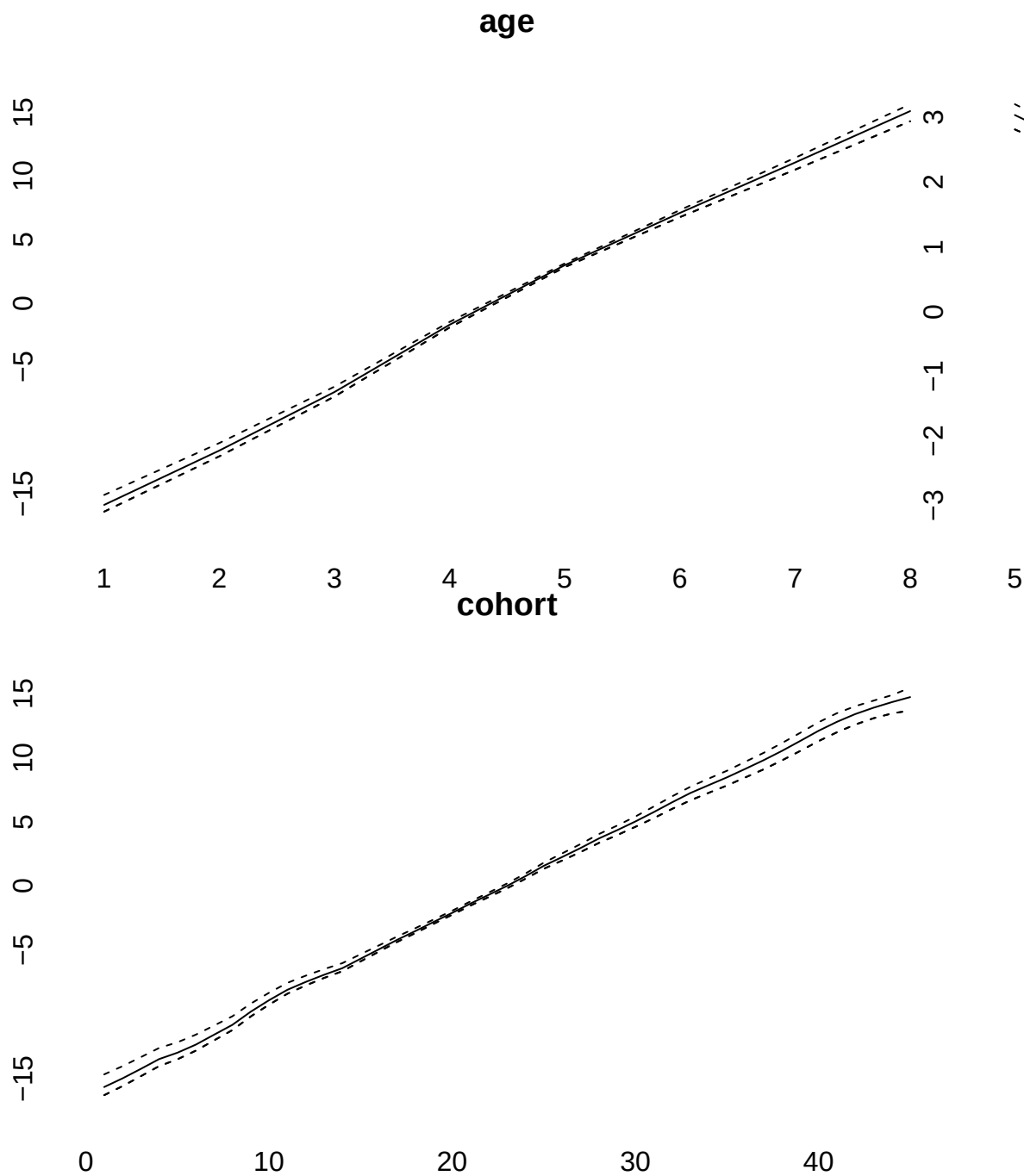
```
checkConvergence(model2)
```

```
## [1] TRUE
```

```
print(model2)
```

```
##
## Model:
## age (rw2) - period (rw2) - cohort (rw2) model
## Deviance:      234.20
## pD:            36.93
## DIC:           271.13
##
##
## Hyper parameters:
## age              5%          50%          95%
## period           1.041       2.935       6.706
## cohort           16.462      41.710      90.368
##                  23.726      44.886      81.580
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
plot(model2)
```

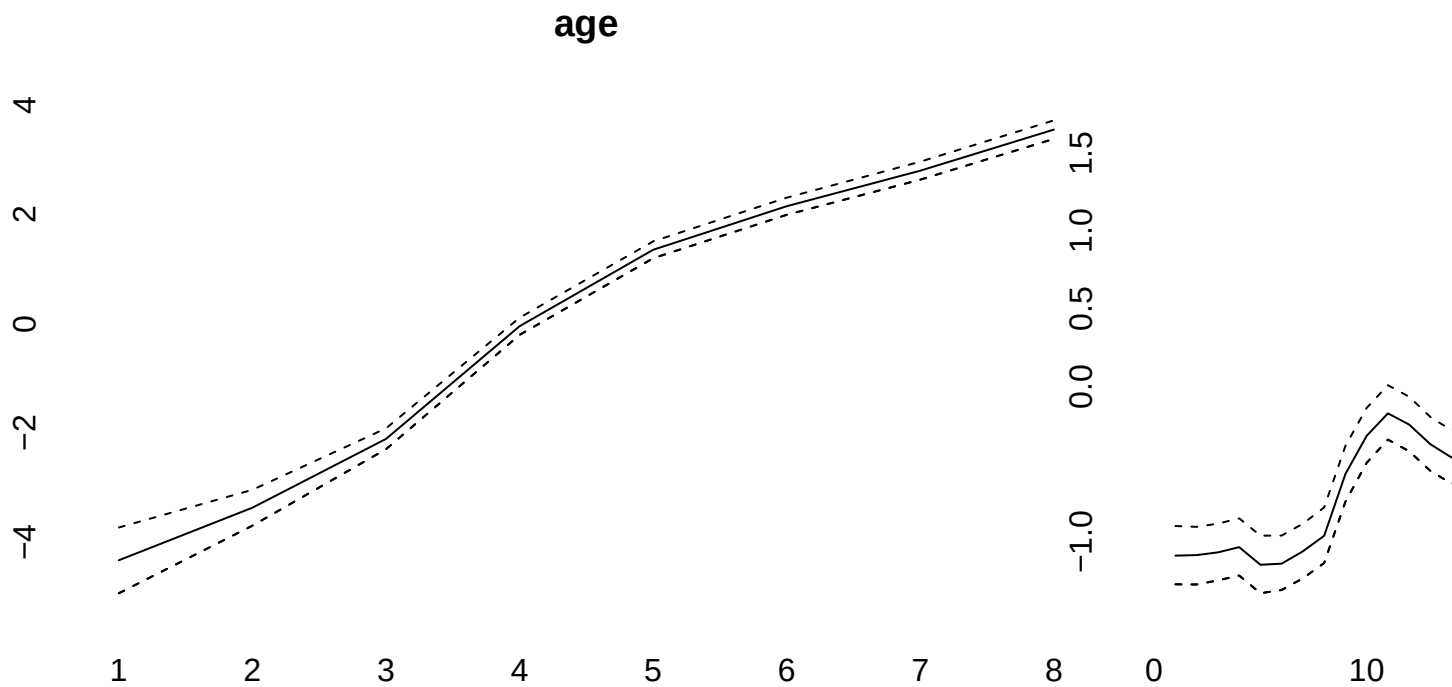


```
model3<-bamp(cases, population, age="rw1", period=" ", cohort="rw2",  
             periods_per_agegroup = 5)  
checkConvergence(model3)  
print(model3)
```

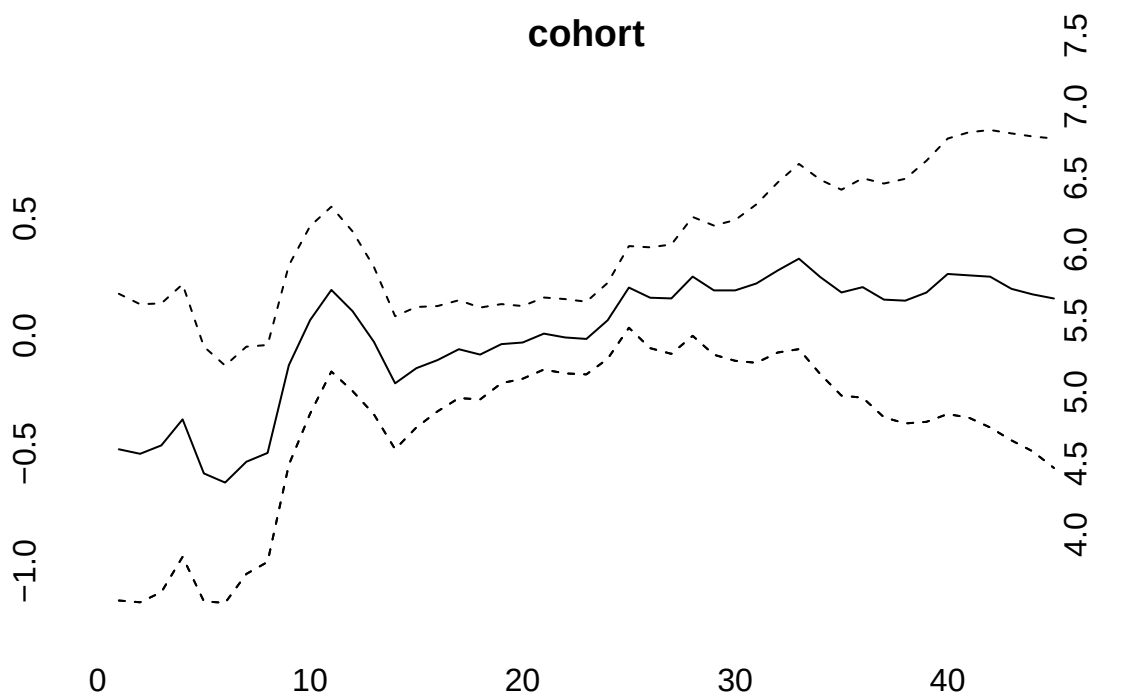
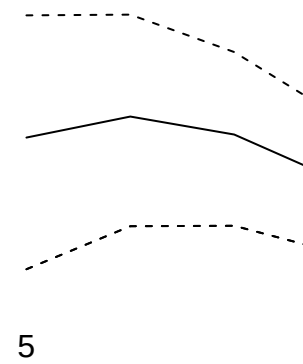
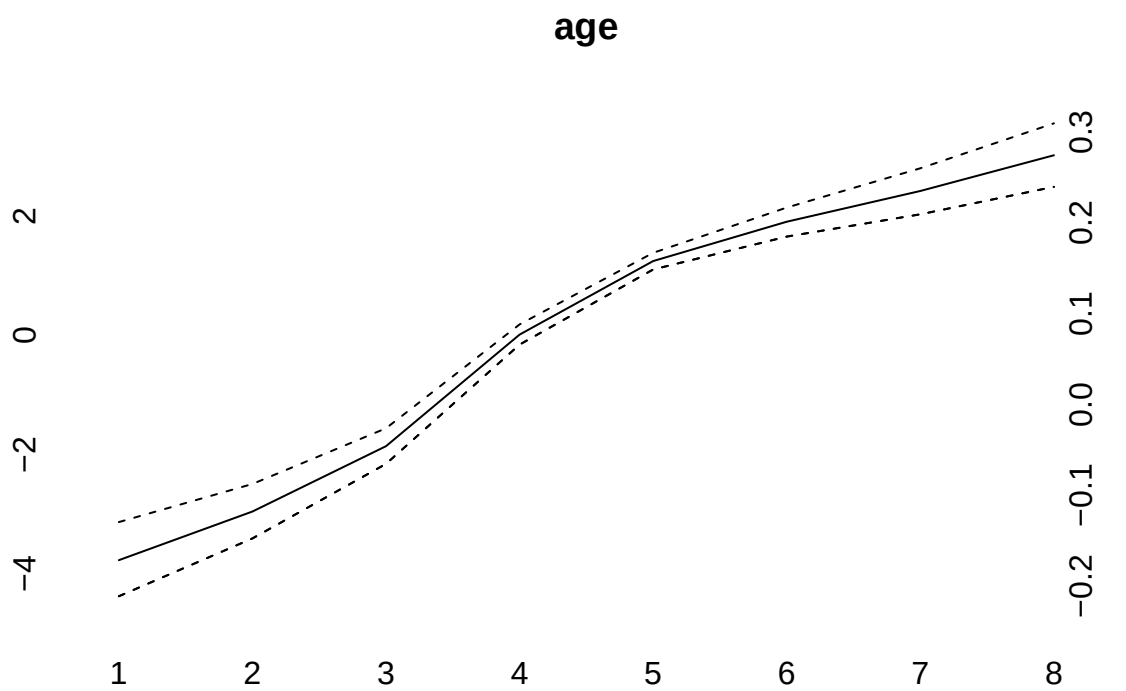
```
## e1  
## Deviance:      276.72
```

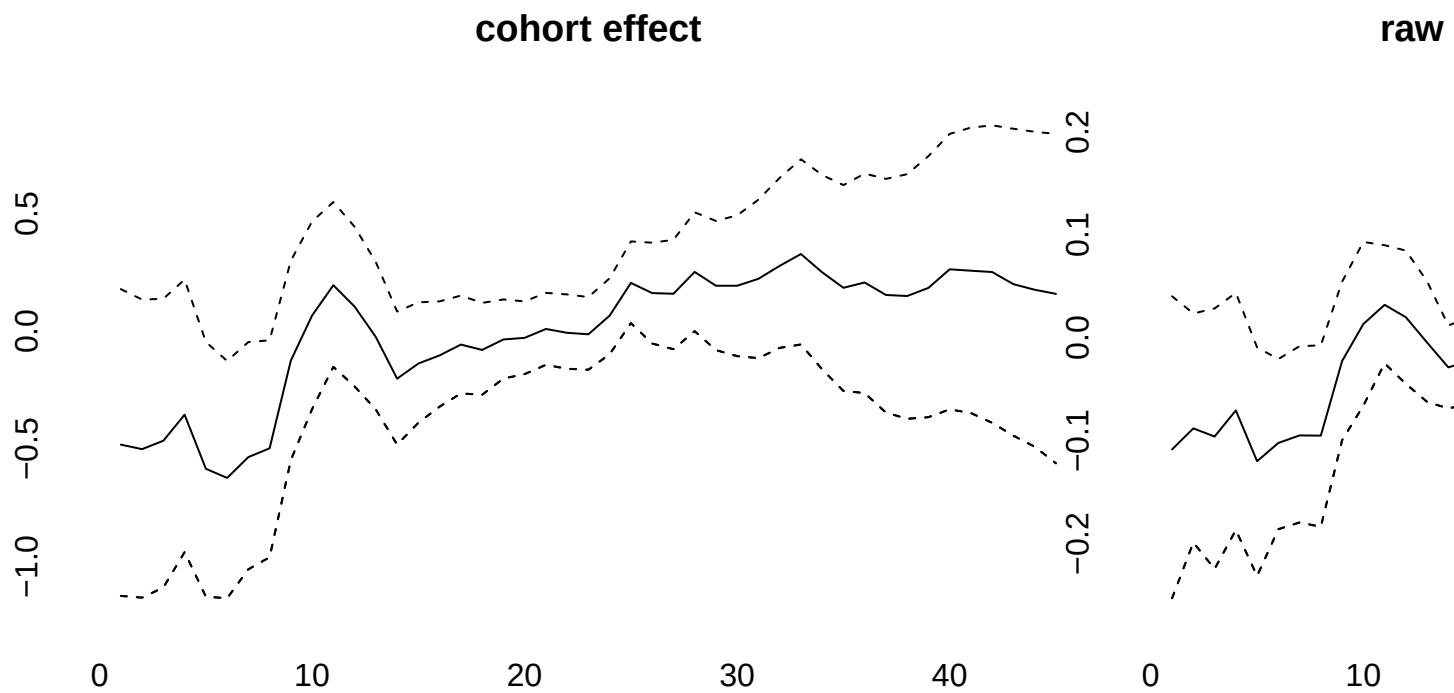
```
## pD:          30.18
## DIC:         306.90
##
##
## Hyper parameters:          5%          50%          95%
## age                       0.288       0.726       1.509
## cohort                    38.002      73.428      139.209
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
plot(model3)
```



```
(model4<-bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
  cohort_covariate = cov_c, periods_per_agegroup = 5))
plot(model4)
```





```
(model15<-bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
               period_covariate = cov_p, periods_per_agegroup = 5))
plot(model15)
```

